



Configuring Layer 3 Switching to Implement Inter-VLAN Routing

Difficulty (out of 3 stars): ★★★

Recommended study time: 2 hours

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1 About This Lab

1.1 Task Description

VLANs are separated at Layer 2 to minimize broadcast domains. To enable the communication between VLANs,

Huawei provides a variety of technologies. The following two technologies are commonly used:

- Dot1q termination sub-interface: Such sub-interfaces are Layer 3 logical interfaces. Similar to a VLANIF interface, after a dot1q termination sub-interface and its IP address are configured, the device adds the corresponding MAC address entry and sets the Layer 3 forwarding flag to implement Layer 3 communication between VLANs. A dot1q termination sub-interface applies to scenarios where a Layer 3 Ethernet interface connects to multiple VLANs.
- VLANIF interface: VLANIF interfaces are Layer 3 logical interfaces. After a VLANIF interface and its IP address are configured, the device adds the MAC address and VID of the VLANIF interface to the MAC address table and sets the Layer 3 forwarding flag of the MAC address entry. When the destination MAC address of a packet matches the entry, the packet is forwarded at Layer 3 to implement Layer 3 communication between VLANs.

In this lab, use VLANIF interfaces to implement inter-VLAN communication. After completing this task, you will be able to understand inter-VLAN communication and be familiar with the method of implementing inter-VLAN communication using VLANIF interfaces.

1.2 Objectives

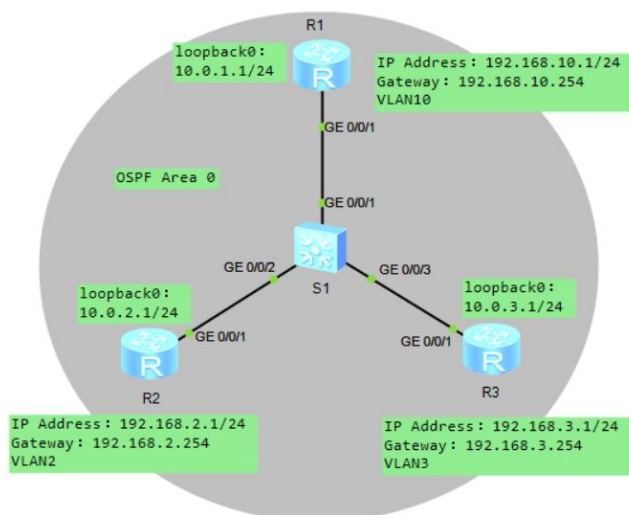
On completing this task, you will be able to:

1. Configure VLANIF interfaces to implement inter-VLAN communication.
2. Understand the forwarding process of inter-VLAN communication.

2 Task Preparation

2.1 Network Topology

Figure 2-1 Lab topology





1. Simulate terminal users on R2 and R3 and assign IP addresses 192.168.2.1/24 and 192.168.3.1/24 to the interfaces.
2. The gateway addresses of R2 and R3 are 192.168.2.254 and 192.168.3.254, respectively.
3. On S1, assign GigabitEthernet0/0/2 and GigabitEthernet0/0/3 to VLAN 2 and VLAN 3, respectively.

2.2 Initial Configuration

- Initial configuration of R1

```
<Huawei>sys
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R1
[R1]int g0/0/1
[R1-GigabitEthernet0/0/1]ip address 192.168.10.1 24
[R1-GigabitEthernet0/0/1]quit
```

- Initial configuration of R2

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R2
[R2]int g0/0/1
[R2-GigabitEthernet0/0/1]ip address 192.168.2.1 24
[R2-GigabitEthernet0/0/1]quit
```

- Initial configuration of R3

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R3
[R3]int g0/0/1
[R3-GigabitEthernet0/0/1]ip address 192.168.3.1 24
[R3-GigabitEthernet0/0/1]quit
```

- Initial configuration of S1

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname S1
[S1]
```

3 Task Implementation

3.1 Checking Basic Configurations

Omitted

3.2 Configuring VLANs on S1

```
[S1]vlan batch 2 3 10
Info: This operation may take a few seconds. Please wait for a moment...done.
```



3.3 Configuring Access Interfaces on S1

```
[S1]int g0/0/1
[S1-GigabitEthernet0/0/1]port link-type access
[S1-GigabitEthernet0/0/1]port default vlan 10
[S1-GigabitEthernet0/0/1]quit
[S1]

[S1]int g0/0/2
[S1-GigabitEthernet0/0/2]port link-type access
[S1-GigabitEthernet0/0/2]port default vlan 2
[S1-GigabitEthernet0/0/2]quit
[S1]

[S1]int g0/0/3
[S1-GigabitEthernet0/0/3]port link-type access
[S1-GigabitEthernet0/0/3]port default vlan 3
[S1-GigabitEthernet0/0/3]quit
[S1]
```

3.4 Configuring VLANIF Interfaces on S1

```
[S1]int Vlanif 2
[S1-Vlanif2]ip address 192.168.2.254 24
[S1] quit

[S1]int Vlanif 3
[S1-Vlanif3]ip address 192.168.3.254 24
[S1-Vlanif3]quit

[S1]int Vlanif 10
[S1-Vlanif10]ip address 192.168.10.254 24
[S1-Vlanif10] quit
[S1]
```

```
[S1]display ip int brief
*down: administratively down
^down: standby
(l): loopback
(s): spoofing
The number of interface that is UP in Physical is 4
The number of interface that is DOWN in Physical is 2
The number of interface that is UP in Protocol is 4
The number of interface that is DOWN in Protocol is 2
```

Interface	IP Address/Mask	Physical	Protocol
MEth0/0/1	unassigned	down	down
NULL0	unassigned	up	up(s)
Vlanif1	unassigned	down	down
Vlanif2	192.168.2.254/24	up	up



Vlanif3	192.168.3.254/24	up	up
Vlanif10	192.168.10.254/24	up	up

[S1]

3.5 Pinging R1, R2, and R3 from S1

```
[S1]ping -c 2 192.168.10.1
PING 192.168.10.1: 56 data bytes, press CTRL_C to break
  Reply from 192.168.10.1: bytes=56 Sequence=1 ttl=255 time=10 ms
  Reply from 192.168.10.1: bytes=56 Sequence=2 ttl=255 time=30 ms

--- 192.168.10.1 ping statistics ---
  2 packet(s) transmitted
  2 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 10/20/30 ms

[S1]ping -c 2 192.168.2.1
PING 192.168.2.1: 56 data bytes, press CTRL_C to break
  Reply from 192.168.2.1: bytes=56 Sequence=1 ttl=255 time=10 ms
  Reply from 192.168.2.1: bytes=56 Sequence=2 ttl=255 time=30 ms

--- 192.168.2.1 ping statistics ---
  2 packet(s) transmitted
  2 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 10/20/30 ms

[S1]ping -c 2 192.168.3.1
PING 192.168.3.1: 56 data bytes, press CTRL_C to break
  Reply from 192.168.3.1: bytes=56 Sequence=1 ttl=255 time=30 ms
  Reply from 192.168.3.1: bytes=56 Sequence=2 ttl=255 time=20 ms

--- 192.168.3.1 ping statistics ---
  2 packet(s) transmitted
  2 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 20/25/30 ms

[S1]
```

3.6 Testing the Connectivity Between VLANs

```
[R2]ping -c 2 192.168.3.1
PING 192.168.3.1: 56 data bytes, press CTRL_C to break
  Reply from 192.168.3.1: bytes=56 Sequence=1 ttl=254 time=40 ms
  Reply from 192.168.3.1: bytes=56 Sequence=2 ttl=254 time=60 ms

--- 192.168.3.1 ping statistics ---
  2 packet(s) transmitted
  2 packet(s) received
```



```
0.00% packet loss
round-trip min/avg/max = 40/50/60 ms
```

```
[R2]
```

```
<R2>tracert 192.168.3.1
traceroute to 192.168.3.1(192.168.3.1), max hops: 30 ,packet length: 40,press CTRL_C to break

 1 192.168.2.254 30 ms  50 ms  50 ms
 2 192.168.3.1 70 ms  60 ms  60 ms
```

VLAN 2 and VLAN 3 can communicate with each other.

3.7 Configuring OSPF

1. Modify the configurations of R1, R2, and R3. Delete the static routes from R1, R2, and R3.

```
[R1]undo ip route-static 0.0.0.0 0.0.0.0 192.168.10.254
```

```
[R2]undo ip route-static 0.0.0.0 0.0.0.0 192.168.2.254
```

```
[R3]undo ip route-static 0.0.0.0 0.0.0.0 192.168.3.254
```

2. Add the loopback0 interface on R1, R2, and R3.

```
[R1]int loopback 0
[R1-LoopBack0]ip address 10.0.1.1 24
[R1-LoopBack0]quit
```

```
[R2]int loopback 0
[R2-LoopBack0]ip address 10.0.2.1 24
[R2-LoopBack0]quit
```

```
[R3]int loopback 0
[R3-LoopBack0]ip address 10.0.3.1 24
[R3-LoopBack0]quit
```

3. Configure OSPF area 0 on R1, R2, R3, and S1.

```
[R1]ospf 1 router-id 10.0.1.1
[R1-ospf-1]area 0
[R1-ospf-1-area-0.0.0.0]network 10.0.1.1 0.0.0.0
[R1-ospf-1-area-0.0.0.0]network 192.168.10.254 0.0.0.0
[R1-ospf-1-area-0.0.0.0]quit
[R1-ospf-1]quit
[R1]
```

```
[R2]ospf 1 router-id 10.0.2.1
[R2-ospf-1]area 0
[R2-ospf-1-area-0.0.0.0]network 10.0.2.1 0.0.0.0
[R2-ospf-1-area-0.0.0.0]network 192.168.2.1 0.0.0.0
[R2-ospf-1-area-0.0.0.0]quit
[R2-ospf-1]quit
[R2]
```



```
[R3]ospf 1 router-id 10.0.1.1
[R3-ospf-1]area 0
[R3-ospf-1-area-0.0.0.0]network 10.0.3.1 0.0.0.0
[R3-ospf-1-area-0.0.0.0]network 192.168.3.1 0.0.0.0
[R3-ospf-1-area-0.0.0.0]quit
[R3-ospf-1]quit
[R3]
```

```
[S1]ospf 1
[S1-ospf-1]area 0
[S1-ospf-1-area-0.0.0.0]network 192.168.2.254 0.0.0.0
[S1-ospf-1-area-0.0.0.0]network 192.168.3.254 0.0.0.0
[S1-ospf-1-area-0.0.0.0]network 192.168.10.254 0.0.0.0
```

4. Check OSPF neighbor relationships on S1.

```
[S1]display ospf peer
```

```
OSPF Process 1 with Router ID 192.168.2.254
Neighbors
```

```
Area 0.0.0.0 interface 192.168.2.254(Vlanif2)'s neighbors
```

```
Router ID: 10.0.2.1      Address: 192.168.2.1
State: Full  Mode:Nbr is Slave  Priority: 1
DR: 192.168.2.1  BDR: 192.168.2.254  MTU: 0
Dead timer due in 34 sec
Retrans timer interval: 5
Neighbor is up for 00:02:20
Authentication Sequence: [ 0 ]
```

```
Neighbors
```

```
Area 0.0.0.0 interface 192.168.3.254(Vlanif3)'s neighbors
```

```
Router ID: 10.0.3.1      Address: 192.168.3.1
State: Full  Mode:Nbr is Slave  Priority: 1
DR: 192.168.3.1  BDR: 192.168.3.254  MTU: 0
Dead timer due in 32 sec
Retrans timer interval: 5
Neighbor is up for 00:02:03
Authentication Sequence: [ 0 ]
```

```
Neighbors
```

```
Area 0.0.0.0 interface 192.168.10.254(Vlanif10)'s neighbors
```

```
Router ID: 10.0.1.1      Address: 192.168.10.1
State: Full  Mode:Nbr is Slave  Priority: 1
DR: 192.168.10.254  BDR: 192.168.10.1  MTU: 0
Dead timer due in 37 sec
Retrans timer interval: 5
Neighbor is up for 00:00:10
Authentication Sequence: [ 0 ]
```

```
[S1]
```



5. Check OSPF routes on R2.

```
<R2>display ospf routing

OSPF Process 1 with Router ID 10.0.2.1
Routing Tables

Routing for Network
Destination      Cost   Type      NextHop        AdvRouter      Area
10.0.2.1/32      0      Stub      10.0.2.1       10.0.2.1       0.0.0.0
192.168.2.0/24   1      Transit   192.168.2.1    10.0.2.1       0.0.0.0
10.0.1.1/32      2      Stub      192.168.2.254  10.0.1.1       0.0.0.0
10.0.3.1/32      2      Stub      192.168.2.254  10.0.3.1       0.0.0.0
192.168.3.0/24   2      Transit   192.168.2.254  10.0.3.1       0.0.0.0
192.168.10.0/24  2      Transit   192.168.2.254  192.168.2.254  0.0.0.0

Total Nets: 6
Intra Area: 6  Inter Area: 0  ASE: 0  NSSA: 0

<R2>
```

6. Ping the loopback0 interfaces of R3 and R1 from the loopback0 interface on R2.

```
<R2>ping -a 10.0.2.1 10.0.3.1
PING 10.0.3.1: 56 data bytes, press CTRL_C to break
Reply from 10.0.3.1: bytes=56 Sequence=1 ttl=254 time=70 ms
Reply from 10.0.3.1: bytes=56 Sequence=2 ttl=254 time=40 ms
Reply from 10.0.3.1: bytes=56 Sequence=3 ttl=254 time=40 ms
Reply from 10.0.3.1: bytes=56 Sequence=4 ttl=254 time=40 ms
Reply from 10.0.3.1: bytes=56 Sequence=5 ttl=254 time=40 ms

--- 10.0.3.1 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 40/46/70 ms

<R2>ping -a 10.0.2.1 10.0.1.1
PING 10.0.1.1: 56 data bytes, press CTRL_C to break
Reply from 10.0.1.1: bytes=56 Sequence=1 ttl=254 time=50 ms
Reply from 10.0.1.1: bytes=56 Sequence=2 ttl=254 time=50 ms
Reply from 10.0.1.1: bytes=56 Sequence=3 ttl=254 time=60 ms
Reply from 10.0.1.1: bytes=56 Sequence=4 ttl=254 time=30 ms
Reply from 10.0.1.1: bytes=56 Sequence=5 ttl=254 time=50 ms

--- 10.0.1.1 ping statistics ---
5 packet(s) transmitted
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 30/48/60 ms

<R2>
```




4 Verification

Omitted.

5 Quiz

1. What configurations must be performed on the switch interface connected to the router when inter-VLAN communication is implemented through sub-interfaces?
2. What changes occur in packets during Layer 3 forwarding?